

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to:

1. _____
2. _____
3. _____

Vocabulary & Key Concepts**Implicit equation:**

Explicitly defined function:

Co-variation:

Rate of change ($\Delta y/\Delta x$):

Hook

“The equation $x^2 + y^2 = 25$ is not a function — but it describes a circle. Can we still talk about how x and y change together? Can we split it into functions?”

Is $x^2 + y^2 = 25$ a function? Why or why not?

Part 1 — Graphing an Implicit Equation

To graph an equation in two variables, _____

Example: $x^2 + y^2 = 25$

Complete the table. (Hint: substitute x and solve for y .)

x	y (positive)	y (negative)
-5	_____	_____
-3	_____	_____
0	_____	_____
3	_____	_____
5	_____	_____

This equation describes _____ explicit function(s):

$y =$ _____ (upper half) $y =$ _____ (lower half)

Solving for one variable in an implicit equation gives _____.

Part 2 — Co-variation

Definition: Co-variation describes _____

Example: Points (3, 4) and (4, 3) are both on the upper half of $x^2 + y^2 = 25$.

$$\frac{\Delta y}{\Delta x} = \frac{\text{_____}}{\text{_____}} = \text{_____}$$

Is this ratio positive or negative? _____

What does this mean about how x and y change together?

Rule:

- If $\Delta y / \Delta x$ is **positive**, the variables _____.
- If $\Delta y / \Delta x$ is **negative**, the variables _____.
- If $\Delta y / \Delta x = 0$, the function has _____ behavior.
- If $\Delta y / \Delta x$ is undefined, the function has _____ behavior.

Part 3 — Second Example: $xy = 4$

Solve for y : $y =$ _____

x	y
1	_____
2	_____
4	_____

Compute $\Delta y / \Delta x$ from $x = 2$ to $x = 4$:

$$\frac{\Delta y}{\Delta x} = \frac{\text{_____}}{\text{_____}} = \text{_____}$$

As x increases, y _____ (increases / decreases). Co-variation is _____ (positive / negative).